NISHANT SHARMA

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Software Engineer with hands-on GenAI engineering experience: LLM integration, RAG pipelines, embeddings, vector stores, guardrails, and agentic workflows. Shipped 2 live AI platforms end-to-end and built secure REST APIs with FastAPI, Spring Boot, and Node.js on Docker, AWS, and CI/CD. Open-source contributor (2 merged PRs) to a 7k+ star LLM engineering repo.

TECHNICAL SKILLS

GenAI & LLM: LLM Integration | Prompt Engineering | RAG | Embeddings | Vector Stores (pgvector) | Agentic Workflows | Guardrails | LLM Cost Monitoring | Whisper | Ollama

Backend & APIs: Python (FastAPI) | Java (Spring Boot, Spring Security) | Node.js | Express.js | REST APIs | JWT | Microservices

Frontend: React.js | Next.js | TypeScript | JavaScript | Tailwind | HTML | CSS

Databases: PostgreSQL | MySQL | pgvector | Prisma ORM | SQL

Cloud & DevSecOps: AWS (EC2, RDS, S3) | Docker | CI/CD (GitHub Actions) | Linux | Secure Coding

Testing & Practices: Unit Testing | Automated Testing | Git | System Design | DSA

WORK EXPERIENCE

AI/ML Engineer | IterativeTech Research, Jaipur **Oct 2026 - Present**

- Building an **agentic QA** platform where **AI** agents generate, run, and report software tests; all **LLM** inference runs locally via **Ollama** so code and test data never leave company infrastructure.
- Developing the **TypeScript**/**Express** backend and **Next.js** dashboard with guardrails: schema-validated JSON output and retry handling for unreliable local-model responses.

Software Engineer Intern | Pratham Software (PSI), Jaipur **Mar 2026 - June 2026**

- Built a 3-layer **Spring Boot** + **MySQL** backend for an employee-appraisal platform with **DTO**-isolated API contracts and global exception handling; secured it with **Spring Security 6** + **JWT RBAC** (HR, Manager, Employee) and input validation blocking privilege escalation.

TECHNICAL PROJECTS

VizTube - Video Platform Backend with AI-Powered RAG | GitHub | Live API **August 2025 - Present**

TypeScript, Node.js, Express, PostgreSQL, pgvector, Prisma, Docker, AWS, FastAPI

- Built 43 **REST** endpoints across 8 modules (auth, uploads, comments, subscriptions, analytics) on a normalized 8-table **PostgreSQL** schema with UUIDv7 keys to prevent **B-tree** fragmentation.
- Engineered a **RAG** pipeline (**Groq Whisper** to 384-dim embeddings to **pgvector**, top-5 chunk retrieval) for video Q&A, running all 3 services on a single 1GB RAM / 2 vCPU **EC2** instance via **Docker CI/CD**.

Weave - Multi-Tenant PDF/CSV RAG System | GitHub **September 2026 - Present**

Python, FastAPI, PostgreSQL, pgvector, Docker, React, TypeScript

- Built a dual-path **RAG** engine (**pgvector** semantic search + generated Text-to-SQL) over 384-dim **embeddings**, eliminating 100% of **LLM** arithmetic hallucination on tabular data.
- Engineered **dual-LLM failover** across 2 providers (**Gemini** to **Groq**) with pre-stream 429 detection for SSE streaming.
- Enforced a **SQL** sandbox blocking 7 mutating keywords (DROP/DELETE/INSERT/UPDATE/ALTER/TRUNCATE/CREATE) with tenant isolation across 5 core tables via collection_id.

OPEN-SOURCE CONTRIBUTIONS

ed-donner/llm_engineering | 7k+ Stars | 2 PRs merged, 1,400+ lines | Repo

- #3631 Built an **LLM cost meter** patching the **OpenAI** client in place to track tokens and spend across 8 providers.
- #3492 Shipped a local **RAG** course navigator (Gradio UI) - offline sentence-transformers embeddings, OpenRouter Q&A.

EDUCATION

- **Bachelor of Computer Applications** | S.S. Jain Subodh PG College, Jaipur |**GPA: 8.3/10.0** | 2024 – 2027
- **Senior Secondary (Grade XII)**| KPS Udaan, Jaipur | **90.2%** |2024

CERTIFICATIONS

- **Microsoft GitHub Foundations** | Verify
- **Back End Development & APIs - FreeCodeCamp** | Verify

ACHIEVEMENTS

- 2024-2025 | **Recommended Twice** | **Indian Air Force** SSB (leadership assessment)
- 2026 | **300+** problems solved | LeetCode, Codeforces, and HackerRank.